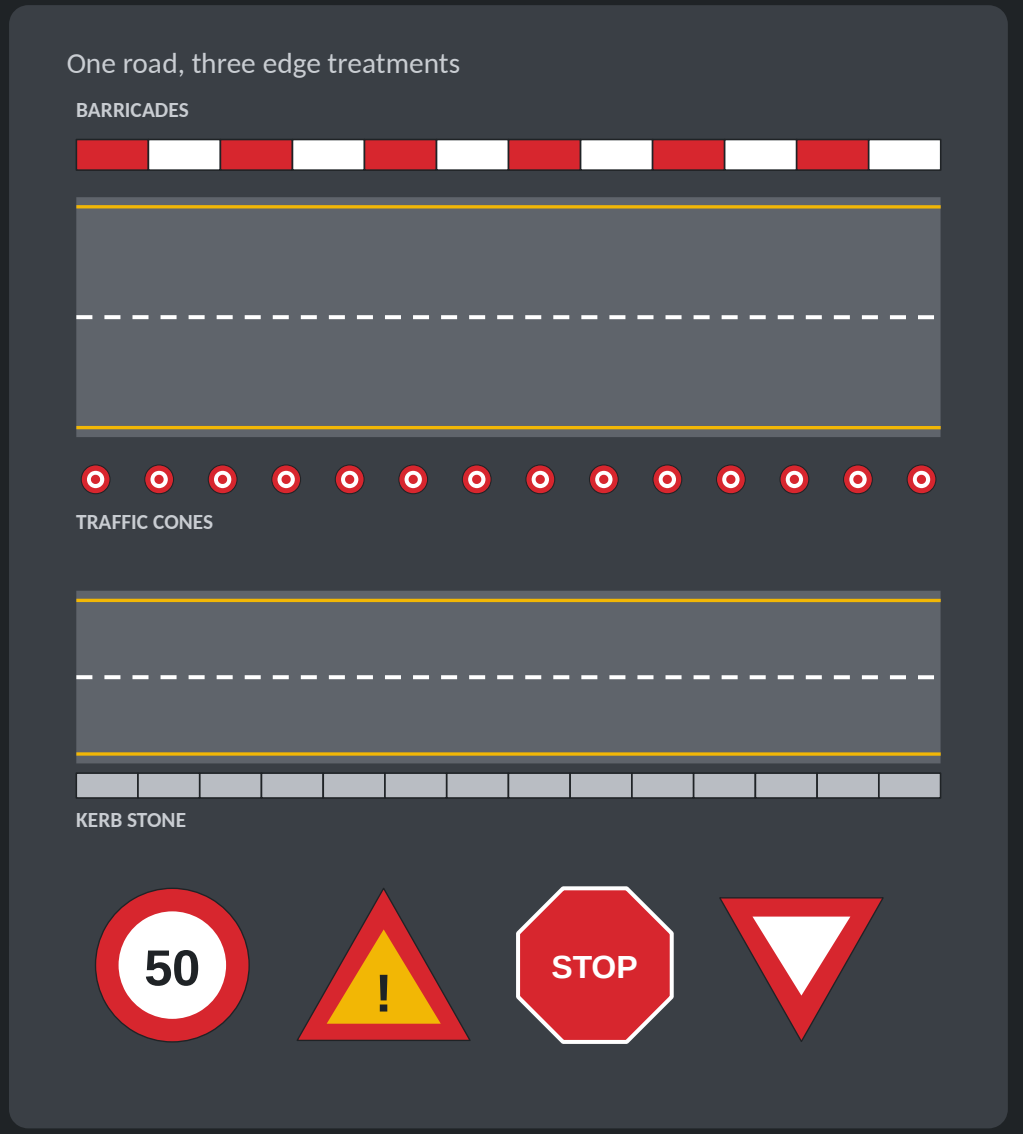
RoadPro

Complete User Manual

Every command and every setting, step by step from start to finish

Version 3.4 • 23 October 2026

Type **RDP** to build a road



Developed by https://www.exceltokml.com/

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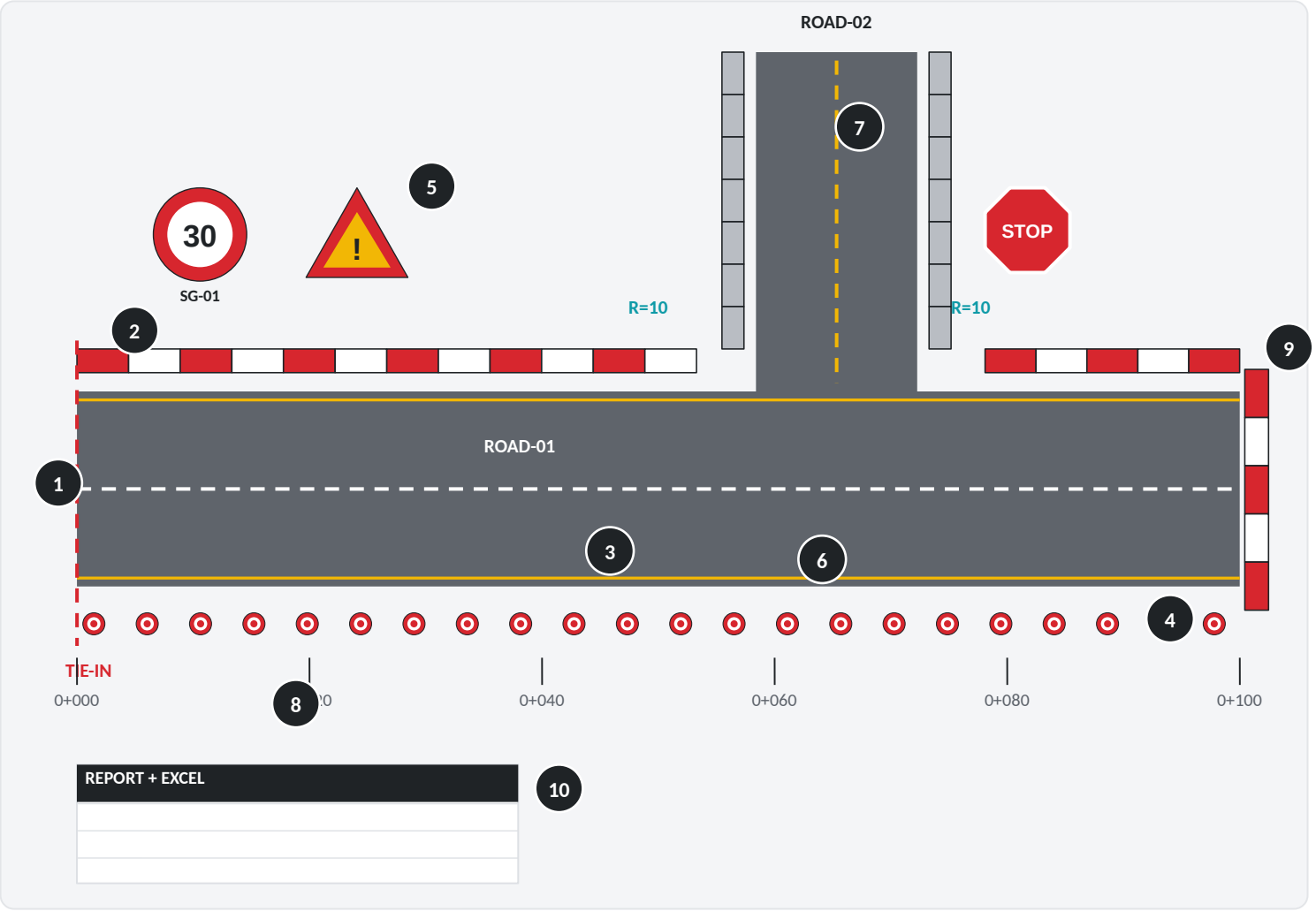
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Lanes, median, sidewalks, walkways, drains, utilities, paver hatch, radius edit, Civil 3D / LandXML, testing

One centre line in, a complete road drawing out



1. **Tie-in or road closed end**
   E / N on the tie-in line, or barricade + ROAD CLOSED board
2. **Edge treatment**
   Barricades, traffic cones or kerb stone, per side
3. **Road surface and markings**
   Hatch, white centre line, yellow L4 edge lines, arrows
4. **Cones / kerb anywhere**
   Same units, also in junction corners and RDL single lines
5. **Signs with IDs**
   Speed, warning, road works, STOP / GIVE WAY, sign schedule
6. **Centre line + radius**
   CL linetype, R, TC / CT, curve data, straights S1, S2
7. **Junction (RDJ)**
   Branch road, kerb returns, corner treatment, aux lane, island
8. **Chainage**
   Ticks and labels at your interval from your start chainage
9. **Widening and taper**
   RDW extra lane, RDT width change, treatment follows
10. **Report**
    Quantities, setting-out E / N, sign schedule, Excel file

What is new in version 3.4

RDP · RDC

Corridor

Lanes (from width, manual or automatic), single or dual carriageway, flush or raised median, sidewalks left / right, walkways, drains and utility corridor (provisions). Slides 38 – 42.

RP_PAVER

Paver hatch

Interlocking I-shape paver hatch on sidewalks and walkways, turning with the road. Herringbone or any other pattern as options. Slide 43.

RDER · RDB

Edit radius later

Change any curve radius after the road is built - the road is rebuilt with its own settings. RDB rebuilds after you edit the CL by hand. Slide 44.

RDLX · RDLXI

Civil 3D / LandXML

Export the CL as a LandXML 1.2 alignment, import LandXML alignments, select a Civil 3D alignment in RDP, alignment checks. Slide 45.

Also fixed / improved: lane widening now moves the edge markings, the report names the real centre marking, RDL no longer prints "cancelled" after a good run, labels stay clear of sidewalks. From 3.3: RDL single line, junction corner treatment.

Upgrading? Load the new RoadPro.lsp with APPLOAD (replace the old file in the Startup Suite). Your settings are kept. Roads drawn with 3.2 / 3.3 work with every 3.4 command; RDER / RDB find their objects by distance and ask first.

Install and load RoadPro

1. **Copy the file to a fixed folder**
   Save RoadPro.lsp in a folder that will not move, for example C:\RoadPro\

2. **Type APPLOAD in AutoCAD**
   Browse to RoadPro.lsp, select it and press Load. Close the dialog.

3. **Check the load message**
   The command line shows Road Pro v3.4 loaded with all the short commands.

4. **Load it every time (recommended)**
   APPLOAD → Startup Suite → Contents → Add → pick RoadPro.lsp → Close.

5. **Security warning?**
   Choose Always Load, or add the folder in OPTIONS → Files → Trusted Locations.

6. **Start working**
   Type RDP and press Enter. The ROAD PRO dialog opens.

```
Command: APPLOAD
  RoadPro.lsp successfully loaded.
Road Pro v3.4 loaded.  ROADPRO (RDP) = build road
  ROADREPORT (RDR) = report  |  ... |
  ROADCORRIDOR (RDC) ... ROADLANDXML (RDLX)
Developed by https://www.exceltokml.com/
```

Requirements

- Full AutoCAD 2014 or newer on Windows (also Civil 3D, Map 3D). AutoCAD LT cannot run LISP.
- Drawing in metres (1 unit = 1 m) or millimetres (1 unit = 1 mm).
- Settings are remembered between sessions on this PC. ROADPRORESET restores the defaults.
- Every command is one UNDO step: type U to take back a whole run.

All commands and short keys

Short	Full command	What it does	You select
RDP	ROADPRO	Build a complete road from a centre line (dialog with all settings)	Pick points, or an existing line
RDR	ROADREPORT	Report on screen, Excel (CSV) file and tables in the drawing	Road centre line
RDRAD	ROADRADIUS	Radius, TC / CT and curve data labels on every curve	Road centre line
RDSEC	ROADSECTIONS	Straight sections S1, S2 … with length, area, chainage, bearing	Road centre line
RDX	ROADXSEC	Cross sections: typical, at one chainage, or at every interval	Road centre line
RDW	ROADWIDEN	Extra lane (widening) with tapers, one side or both sides	Road centre line
RDT	ROADTAPER	Change the road width from a chainage onward	Road centre line
RDJ	ROADJUNCTION	Join a branch road: kerb returns, corner treatment, signs, aux lane, island	MAIN road centre line
RDM	ROADMARK	Draw or remove road markings	Road centre line
RDL	ROADLINE	Single line of barricades, cones or kerb stone on any line	Lines, polylines, arcs, splines
RDC	ROADCORRIDOR	**NEW Lanes, median, sidewalks, walkways, drains, utilities of a road**	Road centre line
RDU	ROADUPDATE	**NEW Regenerate corridor + markings of a road**	Road centre line
RDER	ROADEDITRADIUS	**NEW Change a curve radius after the design, road rebuilt**	Road CL on the curve
RDB	ROADREBUILD	**NEW Rebuild a road after editing its CL by hand**	Road centre line
RDLX	ROADLANDXML	**NEW Export the CL as a LandXML 1.2 alignment (Civil 3D)**	Road CL or polyline
RDLXI	ROADLANDXMLIN	**NEW Import LandXML alignments as polylines**	LandXML file
—	ROADPRORESET	Restore every setting to the default values	—

The workflow from start to finish

1 APPLOAD

Load

Load RoadPro.lsp once, or keep it in the Startup Suite

2 RDP

Build the road

Settings, centre line, then everything is drawn

3 RDRAD

Check curves

Radius labels (and RDSEC straights) if not added in RDP

4 RDW

Widen / taper

Extra lanes (RDW) and width changes (RDT)

5 RDJ

Junctions

Join branch roads with kerb returns and corners

6 RDL

Single lines

Lane closures, work zones, kerb on any line

7 RDM

Markings

Draw or redraw markings when needed

8 RDX

Cross sections

Typical, one chainage or every interval

9 RDR

Report

Run last: quantities, setting-out, Excel

Run RDR last The report reads everything stored on the CL (widenings, tapers, junctions, markings, corridor). Change later: RDC corridor, RDER radius, RDB after editing the CL by hand.

Before you start: prepare the drawing

Checklist

Units. Decide 1 unit = 1 m (default) or 1 unit = 1 mm and set Drawing units in RDP. Values in all dialogs are always typed in metres. RDL uses the same setting.

Direction. Draw the centre line in the chainage direction. The first point is the start chainage.

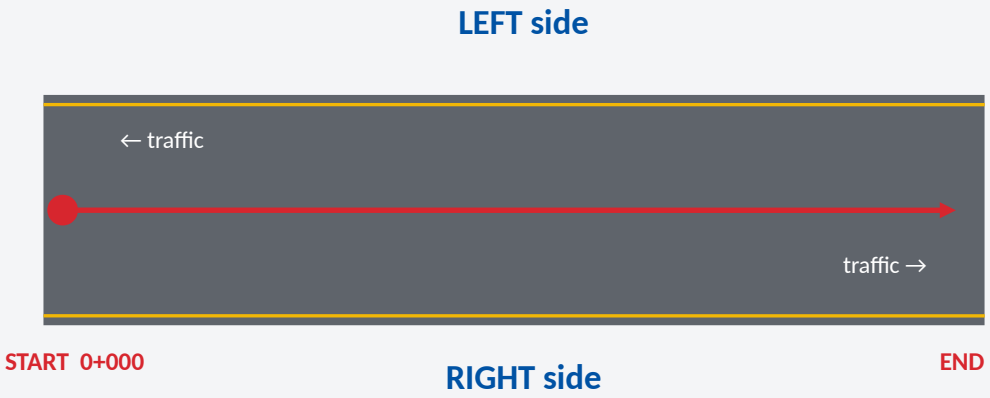
One object. Use one Line, Arc or 2D (LW) Polyline with straight and arc segments. Join pieces first with JOIN; explode blocks.

Survey points. Turn on OSNAP and snap the CL to your design or survey points.

Space and UCS. Work in model space. RoadPro switches to the World UCS while it runs and restores yours.

Save the DWG. Save before RDR: the Excel (CSV) file is written next to the drawing.

Left and right, and chainage



- Left / right are seen looking along the CL from its start point. Traffic keeps right.
- Chainage can be typed as 150 or 0+150, and 1250.5 or 1+250.500.
- Chainages are real road chainages: a road that starts at 1+000 has its widening at 1+150, not at 150.

Build a road step by step

1. **Type RDP and press Enter**
   The ROAD PRO dialog opens with your last settings.

2. **Road Geometry**
   Road name, width, turning radius, units, centre line input, keep original line.

3. **Road Edge and Centre**
   Left / right edge, Barricades, Traffic cones or Kerb stone, centre, sizes, shoulder.

4. **Signs and Markings**
   Speed, warning signs, control at tie-in, sign set.

5. **Surface, Chainage, Ends, Report + Lanes... button**
   Hatch, markings, chainage, end types, report, section; lanes / median / sidewalks (slide 39).

6. **Press OK, then give the centre line**
   Pick points: click points, A = arc, L = line, U = undo, Enter = finish. Select existing: click the line.

7. **RoadPro builds the road**
   Edges, surface, treatment, markings, signs, chainage, radius, ends. Summary printed.

8. **Report and section**
   Excel saved → pick a corner for tables (Enter skips) → open Excel? → pick a point for the typical section (Enter skips).

```
Command: RDP
  (dialog: settings, press OK)
Draw the road centre line
(Arc option allowed, Enter to finish):
Specify start point: pick
Specify next point or [Arc/...]: pick
Specify next point or [Arc/...]: A
Specify end point of arc: pick
Specify end point of arc or [Line]: L
Specify next point: Enter
====== ROAD PRO - ROAD-01 ======
  Road width : 7.30 m ...
  Excel report saved : C:\...\ROAD-01_RoadPro_Report.csv
Pick top-left corner for report tables
<Enter = skip>: pick
Open the Excel report now? [Yes/No] <Yes>: Y
Pick point for the TYPICAL CROSS SECTION
centre line <Enter = skip>: pick
```

The RDP dialog at a glance (default values shown)

ROAD PRO - Road Builder for AutoCAD v3.4

1 Road Geometry

Road name / ID:	**ROAD-01**
Road width (m):	**7.30**
Turning radius:	**Match existing**
Manual radius (m):	**10.0**
Radius measured at:	**Centre line (CL)**
Drawing units:	**Meters**
Centre line input:	**Pick points**

☑ Keep the original line (build on a copy)
☑ Show radius, TC / CT and curve data
☑ Show straight sections (S1, S2…)

2 Road Edge and Centre

☑ Left edge ☑ Right edge

Edge:	**Barricades (red / white)**
Centre:	**Centre line markings**
Unit length / cone spacing (m):	**2.00**
Unit / cone / kerb width (m):	**0.50**
Paved shoulder (m):	**0.50**

3 Signs and Markings

☑ Speed limit	**30 km/h**
☑ Bend ahead before (m)	**30**

☑ Road works ahead (with plate)
☑ Chevron boards on bends
☑ Keep right at median
☑ Lane direction arrows

At tie-in:	**STOP sign**

☑ Two-way traffic (signs both ways)
☑ Sizes and distances by speed (HDS)

Sign set:	**Temporary**
Symbol (m) / Spacing (m):	**2.0 / 15**

4 Surface and Centre Line

☑ Hatch road surface

Hatch pattern / scale:	**SOLID / 1.0**

☑ Show CL text in centre line
☑ Road markings (centre + edge)

Centre marking:	**Standard L1/L2/L2A**

☐ SLOW markings before bends
☑ Delineator posts on curves

5 Chainage

☑ Add chainage ticks and labels

Interval (m):	**20**
Start chainage (m):	**0**
Text height (m):	**1.0**

Panels 1 – 6 : slides 11 – 17

6 Road Ends and Report

Start end:	**Tie-in line**
Finish end:	**Tie-in line**

☑ Report tables + Excel (CSV) file
☑ Typical cross section drawing

Lanes, median, sidewalks, drains, utilities…

Single carriageway, lanes from road width (button opens the corridor dialog, slide 39)

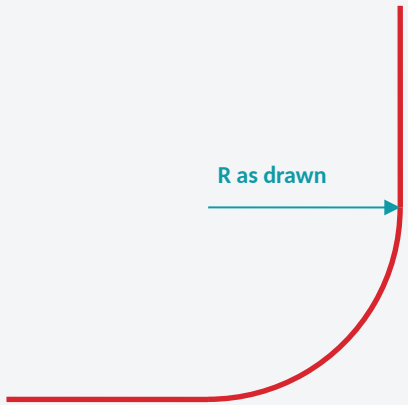
Road Geometry settings

Setting	Options (default)	What it does
Road name / ID	ROAD-01	Name on labels, report title and the Excel file name ROAD-01_RoadPro_Report.csv. Use a new name for each road.
Road width (m)	7.30	Full carriageway width, edge to edge; half on each side of the CL. Paved shoulder is added outside it.
Turning radius	Match existing (keep CL as drawn) / Manual radius at every bend	Match existing keeps your arcs exactly. Manual puts the radius below into every corner of the CL.
Manual radius (m)	10.0	Used only with Manual radius (greyed out otherwise).
Radius measured at	Centre line (CL) / Inner edge (kerb line)	Only with Manual radius. Inner edge: the value is the kerb radius and the CL radius becomes R + half width.
Drawing units	Meters / Millimeters	1 unit = 1 m or 1 unit = 1 mm. Every value in every dialog is still typed in metres.
Centre line input	Pick points (draw new) / Select existing	Draw the CL after OK, or click a line, polyline, arc or Civil 3D alignment.
Keep the original line	On	Select existing only: the road is built on a copy, your line stays as it is.
Show radius, TC / CT and curve data	On	Radius arrow and value, TC / CT ticks with chainage, curve data box (same as RDRAD).
Show straight sections (S1, S2…)	On	Extent bar with length, area, chainage range and bearing for every straight (same as RDSEC).

Tip Using a CL from the designer? Keep Match existing. Drawing a quick site road with sharp corners? Choose Manual radius and RoadPro rounds every bend for you.

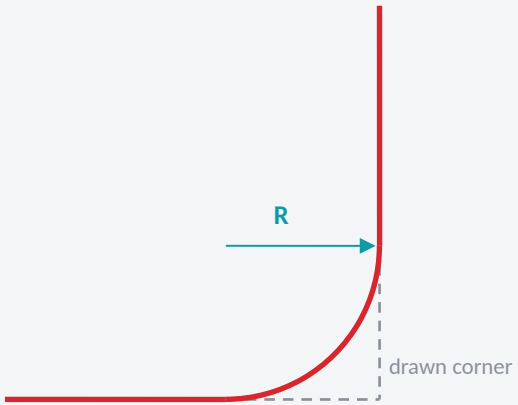
Turning radius: the two options explained

Match existing



Your arcs are kept exactly as drawn. Corners without an arc stay sharp.

Manual radius at every bend



Sharp polyline corners get the Manual radius. Good for quick site roads.

Radius measured at

Width 7.30 m	R typed = 10
At CL: inner edge	6.35 m
At CL: centre line	10.00 m
At CL: outer edge	13.65 m
Inner edge: inner	10.00 m
Inner edge: CL	13.65 m
Inner edge: outer	17.30 m

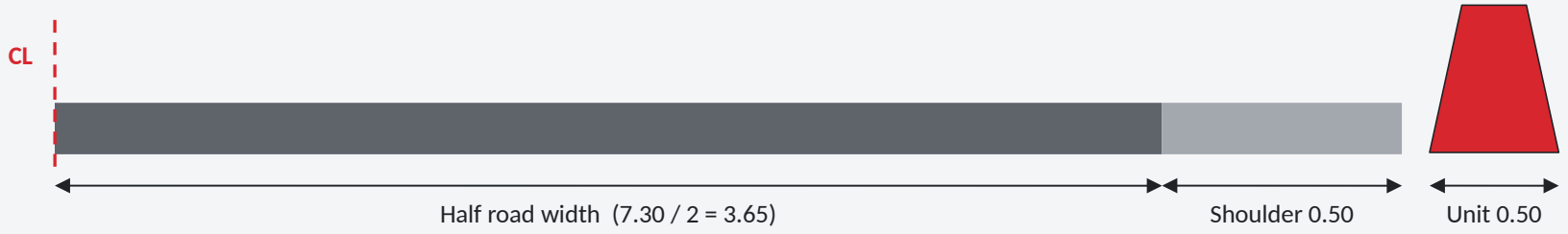
Use Inner edge when the drawing gives a kerb radius.

Rule The CL radius must be larger than half the road width + paved shoulder + unit width, otherwise the inside treatment folds over itself. Example: 7.30 m road, 0.50 shoulder, 0.50 units → CL radius more than 3.65 + 0.50 + 0.50 = 4.65 m. RoadPro prints a warning if a bend is too tight: increase the radius or reduce the width / shoulder.

Road Edge and Centre settings

Setting	Options (default)	What it does
Left edge / Right edge	Both on	Which sides get the edge treatment. Untick one side for a road along a wall, a fence or an existing kerb.
Edge	Barricades (red / white) / Traffic cones / Kerb stone	What is placed along the edges, outside the paved shoulder.
Centre	Centre line markings (no barrier) / Barricades / Traffic cones	Markings only, or a median of barricades or cones on the CL with KEEP RIGHT signs at each nose.
Unit length / cone spacing (m)	2.00	Barricades and kerb: length of one unit. Cones: distance from cone to cone.
Unit / cone / kerb width (m)	0.50	Barricade width, cone base size or kerb width (for kerb use 0.15 – 0.30).
Paved shoulder (m)	0.50	Paved strip between the carriageway edge and the treatment. Hatched, in the section and in the report. 0 = treatment on the edge.

Where each width goes (half section)

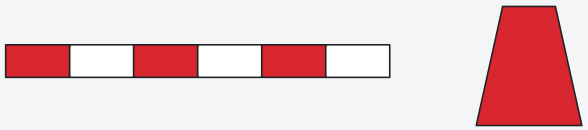


Cones and kerb sit in the same place as barricades. In RDX the section shows the right profile automatically.

HOW TO

How to add barricades, traffic cones or kerb stone

Barricades



1. RDP → Road Edge and Centre
2. Tick Left edge and / or Right edge
3. Edge = Barricades (red / white)
4. Unit length 2.00, width 0.50
5. OK and draw the CL

Red / white units with black outline. Report: red and white units per side. RDX: water-filled or New Jersey profile with its height.

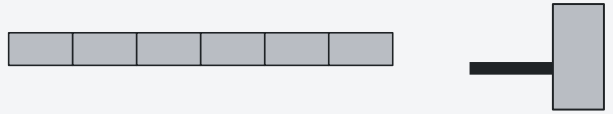
Traffic cones



1. RDP → Road Edge and Centre
2. Tick the sides you want
3. Edge = Traffic cones
4. Unit length = cone spacing, e.g. 3.00
5. Width = cone size, e.g. 0.40

Red cones with a white band (ROAD-CONE, ROAD-CONE-BAND). Report: number of cones per side. RDX: 0.75 m cone.

Kerb stone



1. RDP → Road Edge and Centre
2. Tick the sides you want
3. Edge = Kerb stone
4. Unit length = kerb unit, e.g. 1.00
5. Width = kerb width, e.g. 0.15

Grey kerb units with outline (ROAD-KERB). Report: kerb length in m per side. RDX: kerb with 150 mm upstand.

Median on the centre line

Centre = Barricades or Traffic cones

Treatment follows changes

RDW, RDT and RDJ move or cut it for you

Just one line somewhere?

Use RDL, the single line command (slide 30)

Signs and Markings settings

Setting	Options (default)	What it does
Speed limit km/h	On, 30	Speed limit sign after each entry. The speed also picks HDS sign sizes, warning distances and L1 / L2 patterns.
Bend ahead before (m)	On, 30	Bend warning sign before every curve. With HDS on, the distance comes from the speed (50 – 300 m).
Road works ahead (with plate)	On	Road works triangle with a distance plate at the entry of each direction.
Chevron boards on bends	On	Chevron boards on the outside of every bend.
Keep right at median	On	KEEP RIGHT sign at each nose of a barricade or cone median.
Lane direction arrows	On	White lane arrows on the straights (5.0 m or 7.5 m by speed).
At tie-in	STOP sign / GIVE WAY sign / No sign	Control sign before each tie-in end of the road.
Two-way traffic	On	Signs for both directions. Off = one-way: signs one way and NO ENTRY at the far end.
Sizes and distances by speed (HDS)	On	Plate sizes and warning distances from Oman HDS 2017 Sec. 19 by speed. Off = Symbol and Spacing below.
Sign set	Permanent / Temporary (road works)	Permanent: yellow / black chevrons. Temporary: black / yellow road-works style.
Symbol (m) / Spacing (m)	2.0 / 15	Size of the sign drawing on plan, and the spacing between signs in a group.

Surface and Centre Line settings

Setting	Default	What it does
Hatch road surface	On	Fills the road between the edges.
Hatch pattern	SOLID	Any AutoCAD pattern: AR-CONC, ANSI31 …
Pattern scale (not SOLID)	1.0	Scale for patterns other than SOLID.
Show CL text in centre line	On	Red CL linetype with the letters CL.
Road markings	On	White centre line and yellow L4 edge lines (same as RDM).
Centre marking	Standard L1 / L2 / L2A	See the three options on the right.
SLOW markings before bends	Off	SLOW written on the road before each bend.
Delineator posts on curves	On	Posts on curves at HDS spacing by radius.

Centre marking options

Standard L1 / L2 / L2A by speed



Double continuous white



Single continuous L2A



Yellow L4 edge lines are drawn in all three cases.

Chainage, road ends and report settings

Chainage

Setting	Default	What it does
Add chainage ticks and labels	On	Tick and label at every interval.
Interval (m)	20	Distance between chainage marks.
Start chainage (m)	0	Chainage of the first CL point. Type 1000 or 1+000.
Text height (m)	1.0	Labels, curve data and report tables.

Road Ends and Report

Setting	Default	What it does
Start end / Finish end	Tie-in line	Tie-in line, Road closed or None at each end.
Report tables + Excel (CSV) file	On	Runs the report (RDR) right after the road is built.
Typical cross section drawing	On	Asks for a point to place a typical section (RDX).



Tie-in line (existing road)

Red dashed line across the road with E / N, plus the control sign (STOP / GIVE WAY) set in panel 3.



Road closed (barricade + sign)

Barricade across the full width and a ROAD CLOSED board. Use for dead ends and work limits.



None

The road simply ends. Use when another road or drawing continues it.

Everything on its own layer

Layer	Contents
ROAD-CL	Centre line, red CL linetype (holds the road data)
ROAD-EDGE	Left and right carriageway edges
ROAD-SURFACE	Surface hatch
ROAD-SHOULDER	Paved shoulder hatch
ROAD-BARRIER-RED / -WHITE	Barricade unit fills
ROAD-BARRIER-OUTLINE	Black outline of units
ROAD-CONE / ROAD-CONE-BAND	Traffic cones and white band
ROAD-KERB	Kerb stone units
ROAD-CHAINAGE	Chainage ticks and labels
ROAD-CURVE	Radius, TC / CT, curve data
ROAD-SECTION	Straights S1, S2 ...
ROAD-MEDIAN (-KERB)	Median fill / hatch and kerbs (3.4)
ROAD-SIDEWALK	Sidewalks, paver hatch (3.4)
ROAD-WALKWAY	Walkways, paver hatch (3.4)

Layer	Contents
ROAD-SIGN (-RED -WHITE -YELLOW -BLUE)	Posts, plates and symbols
ROAD-MARKING (-FILL)	White lines, arrows, text
ROAD-MARKING-YELLOW	Yellow L4 edge lines
ROAD-DELINEATOR	Delineator posts on curves
ROAD-TIEIN	Tie-in line and E / N
ROAD-WIDENING	Widening and taper labels
ROAD-JUNCTION	Junction labels, corner R, RDL labels
ROAD-ISLAND	Splitter islands
ROAD-REPORT	Report tables
ROAD-XSEC ...	Cross section parts and text
ROAD-01, ROAD-02 ...	Road name groups
ROAD-DRAIN	Drainage provision (3.4)
ROAD-UTILITY	Utility corridor provision (3.4)
ROAD-LANDXML	Imported LandXML alignments (3.4)

Tip Freeze or turn off layers for clean plots (e.g. ROAD-CHAINAGE). Black outlines look white on a dark screen but plot black.

Radius and straight sections on the plan

RDRAD — Every curve

```
C1
R inner = 10.000 m
R CL    = 13.650 m
R outer = 17.300 m
L (CL)  = 21.441 m
A       = 156.52 m²
Δ       = 90°00'00"
```

Radius line with arrow from the centre point, R on the line, TC and CT ticks across the road with chainage. Layer ROAD-CURVE.

RDSEC — Every straight

```
S1  STRAIGHT
L   = 45.230 m
A   = 330.18 m²
CH  0+012.000 - 0+057.230
BRG 45°12'30"
```

Extent bar with end ticks, label parallel to the road. Bearing is whole-circle from North. Layer ROAD-SECTION.

1. **Type RDRAD (or RDSEC)**
   Select the road centre line on layer ROAD-CL.

2. **Labels are added**
   The command line says how many curves / straights were labelled.

When to use If you switched the labels off in RDP, deleted them, or changed the road with RDW / RDT. Erase the old labels on ROAD-CURVE / ROAD-SECTION first so they are not doubled.

Signage placed automatically, traffic keeps right



Road works ahead

Entry of each direction



Speed limit

One spacing after entry



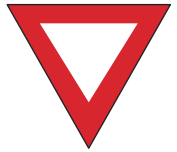
Bend ahead

Distance by speed



STOP

Before each tie-in



Give way

Option at tie-ins



No entry

Far end, one-way road



Keep right

Median nose

How each sign is drawn on the plan



SG-01

Post and plate. Black dot outside the edge treatment; plate faces the traffic.

Leader, symbol and ID. IDs SG-01, SG-02 … follow the chainage order.

Sizes. By speed (HDS) or your Symbol size.

Also drawn and reported

- Chevron boards on the outside of bends (permanent yellow / black, temporary black / yellow)
- ROAD CLOSED boards at closed ends
- White lane arrows on the straights, delineator posts on curves
- STOP / GIVE WAY on every junction branch (RDJ)
- Sign schedule in the report: ID, sign, plate size, chainage, side, facing, post E / N

Road markings: draw, remove and quantities

1. **Type RDM**
   Select the road centre line.

2. **Choose Draw or Remove**
   Road markings [Draw/Remove] <Draw>: press Enter to draw.

3. **Markings are drawn**
   Centre line by your Centre marking setting, yellow L4 edge lines, bend warnings, arrows, SLOW (if on).

4. **Remove**
   Deletes the markings RDM drew and clears their quantities from the report.

Good to know New roads: tick Road markings in RDP and you do not need RDM. RDJ redraws the markings of both roads with the junction gaps and adds turning-area markings.

Line	Use	Pattern / size
L1	Lane / centre line	2/6, 3/9, 4/12 m by speed, white
L2	Warning before bends	6/2, 9/3, 12/4 m
L2A	No overtaking, bends, approaches	Continuous
L4	Edge line	Yellow, 120 mm (150 mm over 100 km/h)
L5	Beside an auxiliary lane	1 m / 1 m
L5/3	Across a side road mouth	600 / 300 mm
L7	Stop line	Solid line + STOP text (2.1 m)
L8	Give way	Double broken line + 3.75 x 1.25 m triangle
Arrows	Lane direction	5.0 m or 7.5 m by speed
Report	Quantities	Length per line type and paint area in m²

Cross sections with widths, layers and heights

1 Type RDX
Select the road centre line.

2 Fill the Cross Section dialog
Choose what to draw and set the values (table).

3 Pick the placement point
Typical / one chainage: point for the CL at finished level. Every interval: top-left corner of the sheet.

4 Sections are drawn
Every interval: 3 per row, up to 60 sections. The note says curve or straight.



Shows edge treatment, and (3.4) lanes, median, sidewalks, drains, utilities

Setting	Options (default)	Notes
Draw	Typical / At one chainage / At every chainage interval	Typical = one generic section
Chainage (m)	0	For At one chainage; 0+150 allowed
Interval (m)	20	For every interval
Crossfall (%)	2.5	Carriageway slope
Crossfall type	Camber / One-way falls right / falls left	Camber falls both sides
Verge beyond barricade (m)	1.0	Ground shown outside
Pavement layers 1 – 3	Asphalt 0.05 / Base 0.20 / Sub-base 0.15	Name + thickness; 0 = not used
Barricade profile	Water-filled / New Jersey, height 1.00	Used when the edge is barricades
Sign post	On, clearance 1.50, sign 0.90	Post with height to the plate

Lane widening step by step

1. **Type RDW, select the road CL**
   The dialog shows the road chainage range and width.

2. **Side and size**
   Side Left / Right / Both, Extra width 3.65 m.

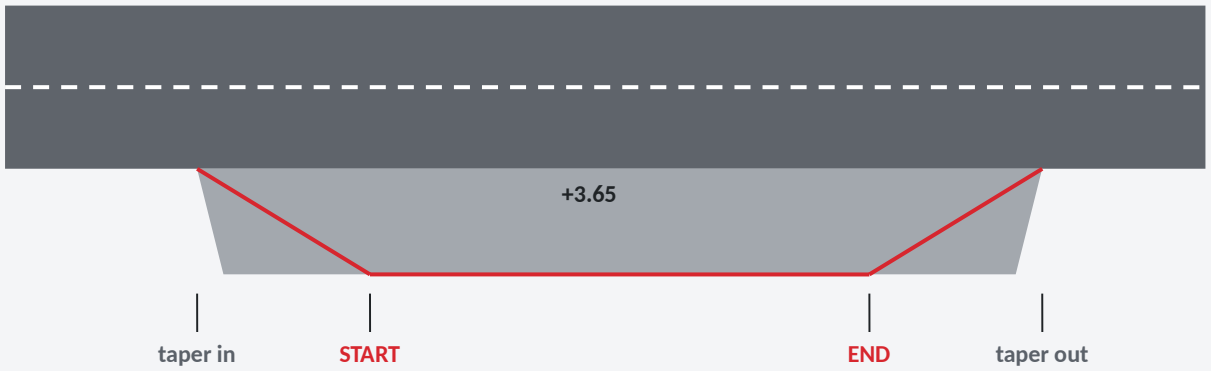
3. **Where**
   Start and End chainage of the full-width lane, Taper in 30 m, Taper out 30 m.

4. **Shape**
   Taper type straight or curve; Match point fillet (radius 30 m) or direct.

5. **Options and OK**
   Move barricades (cones and kerb too), lane arrow, labels and dimension.

6. **Result**
   New edge drawn, old edge trimmed, treatment moved, markings and report updated.

Right side widening



Setting	Default	Notes
Side	Right	Left / Right / Both sides
Extra width (m)	3.65	Width of the new lane
Start / End chainage	100 / 200	Full-width part, real chainage
Taper in / out (m)	30 / 30	Before start and after end
Fillet radius (m)	30	Round match points

Both sides, different chainages, taper shapes and rules

Both sides with different chainage

- Set Side = Both sides. The top block becomes the LEFT side.
- Tick Different chainage on right side: the right gets its own width, start, end, tapers and fillet radius.
- Untick it and both sides use the same values.

Taper type and match point

Straight (linear) taper: straight line from old to new edge.

Curve (reverse-curve) taper: smooth S-shape.

Fillet at match points: rounds the joins with the fillet radius (labelled R=30).

Direct link: sharp joins, no fillet.

Rules that keep it inside the road

- Start – taper in must be at or after the road start.
- End + taper out must be at or before the road end.
- End must be larger than Start.
- Keep junctions away from widened or tapered edges.

Options at the bottom of the dialog

Move barricades to the new edge: barricades, cones or kerb follow the new edge.

Lane arrow in the new lane: white arrow in the widened lane.

Labels, width dimension, match points: text on ROAD-WIDENING with chainages.

Road taper: change the width from a chainage onward

1. **Type RDT, select the road CL**
   The dialog shows the chainage range and the design width.

2. **Side**
   Left, Right or Both sides (default Both).

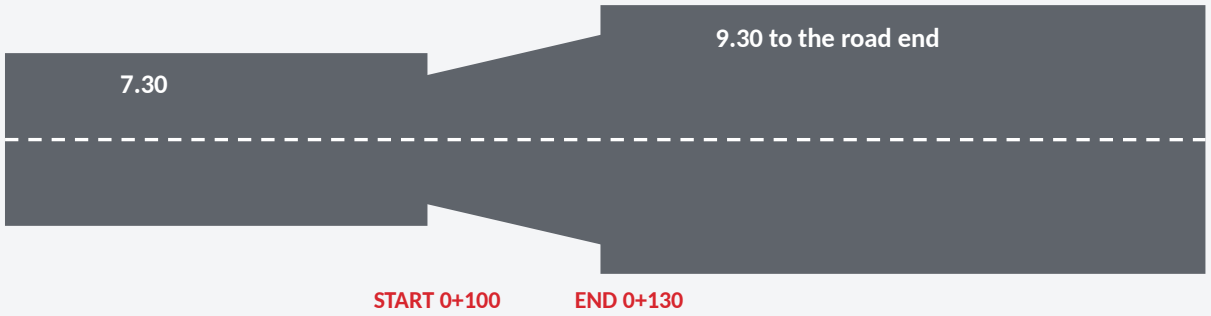
3. **Taper start and end chainage**
   Default 100 and 130: the width changes over these 30 m.

4. **Edge shift per side (m)**
   + = wider road, – = narrower road. +1.00 both sides makes 7.30 → 9.30 m.

5. **Shape and options, OK**
   Taper type, match point, fillet radius, barricades follow, labels.

More tapers? The new width continues to the road end. Run RDT again at another chainage to change it again (e.g. back to 7.30).

Edge shift +1.00 m both sides



Setting	Default	Notes
Edge shift per side (m)	+1.00	– values narrow the road
Different chainage on right	Off	Own start, end, shift, radius
Taper type / match point	Straight / fillet	Fillet radius 30 m
Barricades follow new edge	On	Cones and kerb too
Labels and match points	On	On ROAD-WIDENING

Junction / tie-in step by step

1. **Build the main road first with RDP**
   The branch uses your RDP settings for barricades, signs and chainage.

2. **Type RDJ, select the MAIN road CL**
   The Road Junction dialog opens and shows the main road range and width.

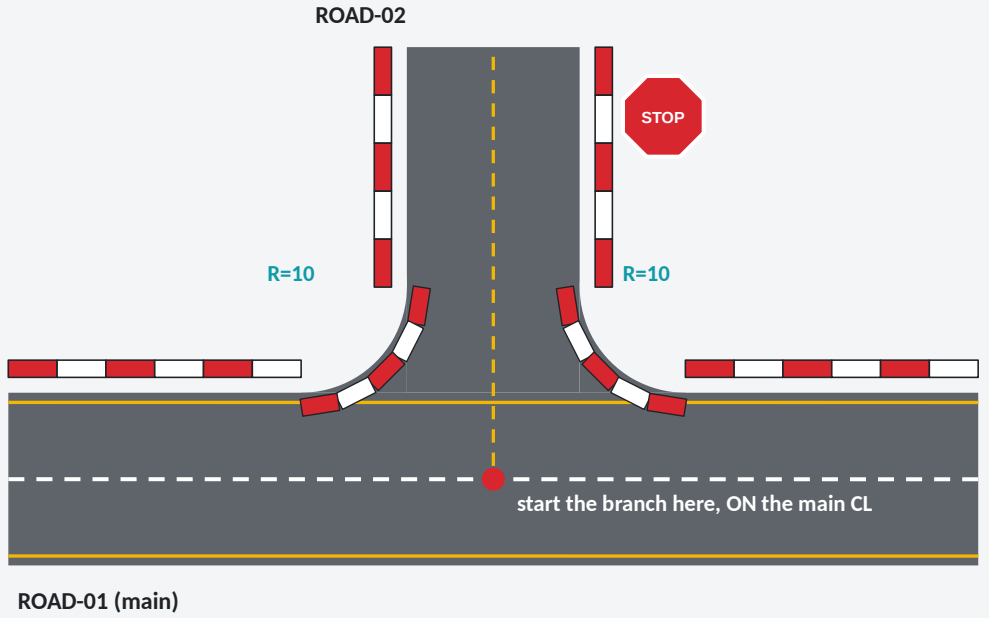
3. **Joining road (branch)**
   Branch name ROAD-02, width 7.30, Branch centre line: Draw new or Select existing.

4. **Junction settings**
   Corner radius 10, control STOP / GIVE WAY, corner treatment, turning-area markings.

5. **Aux lane and island, then OK**
   Auxiliary lane on the main road, splitter island on the branch.

6. **Draw the branch CL**
   First point ON the main road CL (snap Nearest), then draw outward and press Enter.

7. **Junction is built**
   Branch road, main edges cut, kerb returns, corners, signs, markings, summary printed.



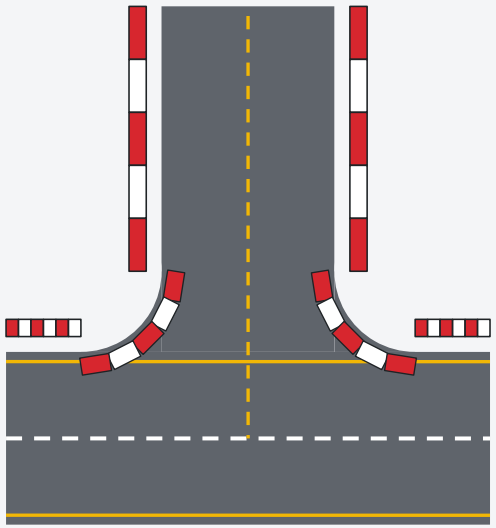
Crossroads? Run RDJ once for each side at the same chainage. Several branches (J1, J2 ...) can join the same main road.

Road Junction dialog: every setting

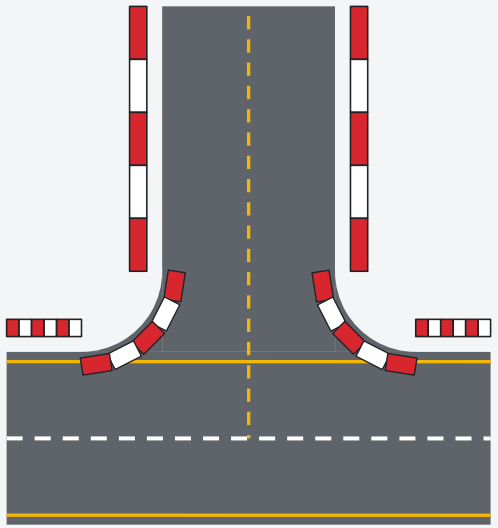
Setting	Options (default)	What it does
Branch road name / ID	ROAD-02	Name of the joining road (labels, report).
Branch road width (m)	7.30	Carriageway width of the branch.
Branch centre line	Draw new (from main road CL) / Select existing	Draw after OK, or click a line that starts on the main CL.
Corner (kerb return) radius (m)	10.0	Radius of both kerb returns.
Control on the branch	STOP / GIVE WAY / No sign	Sign and L7 stop line or L8 give-way line on the branch.
Edge treatment around the corners	On	Places a treatment along both kerb returns.
Corner treatment	Same as main road / Barricades / Traffic cones / Kerb stone	**NEW in 3.3: choose what goes around the corners (slide 28).**
Road markings in the turning area	On	L5/3 across the mouth, L2A approach, turn arrow, STOP / triangle.
Add auxiliary lane (two taper lines)	On	Extra lane on the main road at the junction.
Lane width / long / short / storage	3.65 / 60 / 30 / 30 m	Width, approach taper (long), departure taper (short), storage length.
Island / width / length	No island / Kerbed / Painted, 1.5 / 15 m	Splitter island on the branch.

How to add barricades, cones or kerb stone at junction corners

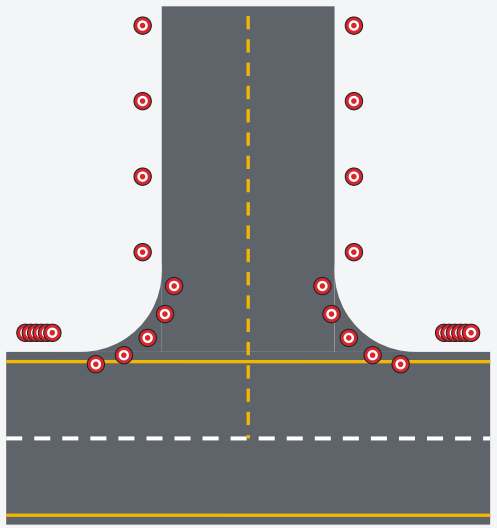
Same as main road



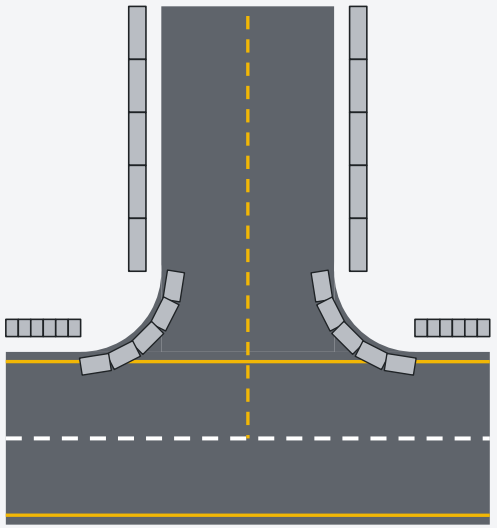
Barricades



Traffic cones



Kerb stone



1. **RDJ → select main CL**
   Open the Road Junction dialog.

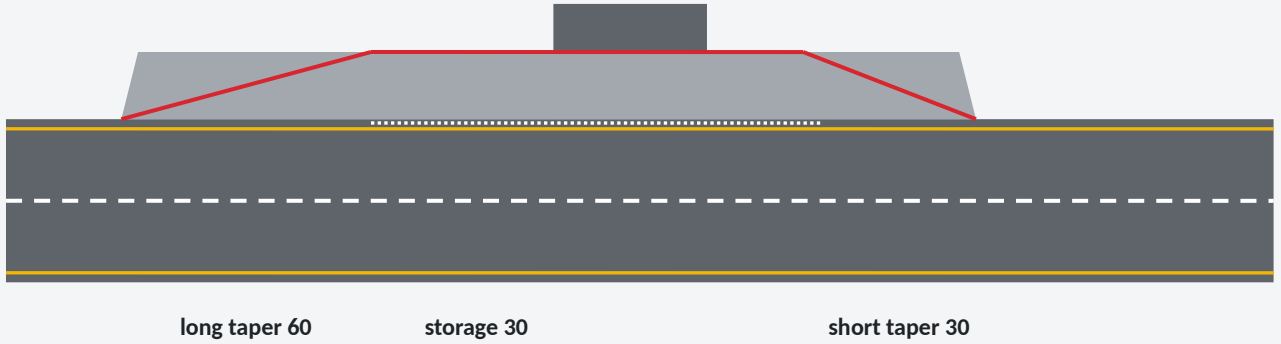
2. **Tick Edge treatment around the corners**
   Untick to leave the corners open.

3. **Choose Corner treatment and press OK**
   Same as main road uses the main road Edge; size = RDP Unit length / cone spacing and width.

```
======= ROAD PRO - JUNCTION J1 =======
  Branch      : ROAD-02  width 7.30 m
  Corner R    : 10.00 m
  Corners     : 14 traffic cones
                around the corners
  Control     : STOP sign
```

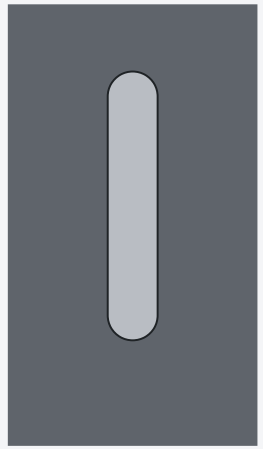
Auxiliary lane and splitter island

Auxiliary lane on the main road



L5 1 m / 1 m beside the lane

Splitter island on the branch



1.5 x 15 m

Kerbed or painted

Setting	Default	Range used	Notes
Lane width (m)	3.65	3.0 – 3.65	Width of the auxiliary lane
Approach taper, long (m)	60	by speed	Entry into the lane, before the junction
Departure taper, short (m)	30	by speed	Exit back to the main road
Storage length (m)	30	by traffic	Full-width part at the junction
Island width / length (m)	1.5 / 15	—	Lanes beside the island under 2.75 m give a warning

Single line of barricades, cones or kerb stone

Use RDL when you need one run anywhere, without building a road:

- Lane closure or cone taper on an existing road
- Work zone or excavation boundary
- Kerb along a footpath, parking or island edge
- Barricades on both sides of a trench or a temporary path
- A ring of cones round a manhole (circle)

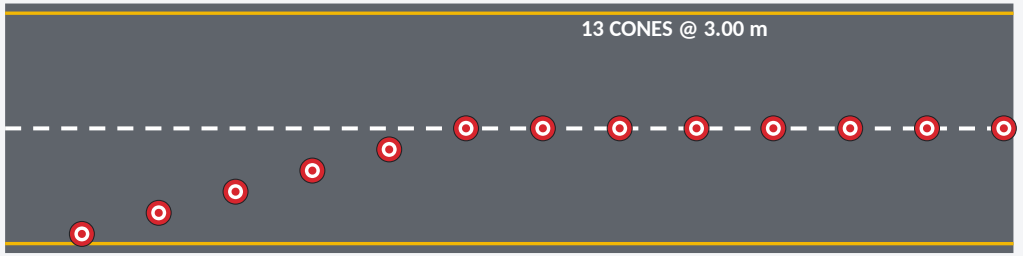
1 **Draw the line**
Line, polyline, arc, spline, circle or ellipse on any layer, in the run direction.

2 **Type RDL and select**
Select one or more objects, press Enter. The dialog opens.

3 **Choose type, placement and size**
Barricades / Traffic cones / Kerb stone; on the line, left, right or both.

4 **Press OK**
Units are placed, the quantity label is added, totals print. U undoes the run.

Example: cone taper closing a lane



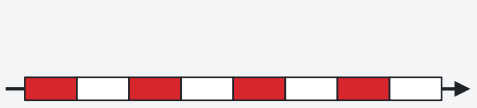
Draw a polyline along the closure, RDL → Traffic cones, On the line, spacing 3.00.

```
Command: RDL
Select lines, polylines, arcs or splines
for the single line run: 1 found, Enter
====== ROAD PRO - SINGLE LINE ======
  Traffic cones   : 13
  Type U to undo this run in one step.
```

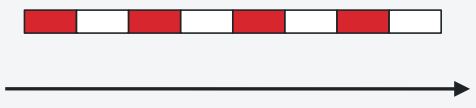
Single Line dialog: every setting

Setting	Options (default)	What it does
Type	Barricades (red / white) / Traffic cones / Kerb stone	What to place along the selected objects.
Placement	On the line (centred) / Offset to the left / Offset to the right / Both sides	Left / right are seen looking along the line from its start point.
Offset from the line (m)	0.50	Distance from the line to the units for left, right and both sides.
Unit length / cone spacing (m)	2.00	Length of a barricade or kerb unit, or spacing of cones.
Unit / cone / kerb width (m)	0.50	Width of the unit or size of the cone.
From distance along line (m)	0	Start the run part-way along the line.
To distance (m, 0 = line end)	0	Stop the run before the line end.
Quantity label on the drawing	On	e.g. 24 BARRICADES (12 R / 12 W), 13 TRAFFIC CONES or 48.0 m KERB STONE.

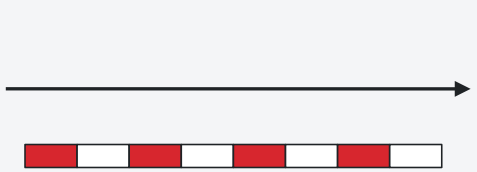
On the line



Offset left



Offset right



Both sides



Make the report and the Excel file

1. **Save the drawing**
   The Excel (CSV) file is written in the same folder as the DWG.

2. **Type RDR, select the road CL**
   Any RoadPro road: main road or branch.

3. **Read the summary**
   Printed on the command line (F2 shows it all).

4. **Excel file is saved**
   ROAD-01_RoadPro_Report.csv next to the DWG.

5. **Place the tables**
   Pick the top-left corner, or Enter to skip the drawing tables.

6. **Open Excel?**
   Yes opens the CSV, No continues. Close it before you run RDR again.

```
Command: RDR
Select the road centre line: pick
======= ROAD PRO REPORT - ROAD-01 =======
  Centre line length : 412.350 m
  Road surface area  : 3010.16 m2
  Edge treatment     : Barricades
  Barricade units    : 418 (209 red / 209 white)
  Signs              : 14
  Excel report saved : D:\Project\ROAD-01_...csv
Pick top-left corner for report tables: pick
Open the Excel report now? [Yes/No]: Y
```

Remember Run RDR after the last change (RDW, RDT, RDJ, RDM) so every quantity is up to date. Enter at the table prompt only skips the drawing tables, never the Excel file.

What is inside the report

Summary

Road name, date, units, start / end chainage, length, width, edge lengths, surface area, straights and curves, design radius, minimum radius

Edge treatment

Barricade units red / white per side, traffic cones per side (nos), kerb stone per side (m), centre median, unit size, paved shoulder and area

Signs and markings

Signs, chevrons, lane arrows, delineators, sign set; marking lengths per line type and paint area (m^2)

Curve data

Each curve: TC / CT chainage, radius, arc length, deflection, centre and TC / CT E / N

Road sections

Every straight and curve: chainage range, length, edge lengths, area, bearing

Setting-out

E / N of CL, left edge and right edge at every chainage interval

Sign schedule

ID, sign, plate size, chainage, side, facing direction, post E / N

Widening, taper, junction, corridor

Extra areas, new edges, kerb returns E / N; lanes, median, sidewalks, walkways, drains, utilities (length, area)

From a blank drawing to a finished road with a junction

1 **APPLOAD** Load RoadPro.lsp. Save the drawing as SITE-ROAD.dwg.

2 **RDP** Geometry: ROAD-01, width 7.30, Match existing, Meters, Pick points.

3 **RDP** Edge: Left + Right, Barricades, 2.00 x 0.50, shoulder 0.50. Centre: markings.

4 **RDP** Signs: 50 km/h, HDS on, Temporary, STOP at tie-in. Chainage 20 from 0.

5 **RDP** Ends: start Tie-in, finish Road closed. Report and section on. OK.

6 **RDP** Draw the CL: 3 straights and an arc, Enter. Place tables, open Excel, place section.

7 **RDW** Right side, 3.65 m, 0+100 to 0+180, tapers 30 / 30, fillet 30.

8 **RDJ** ROAD-02, 7.30 m, R 10, STOP, corners Traffic cones, aux lane on. Branch from 0+300.

9 **RDL** Polyline round an excavation: Barricades, Both sides, offset 1.00.

10 **RDX** At every chainage interval, 20 m: pick the sheet corner.

11 **RDR** Final report: run on ROAD-01 and on ROAD-02.

Result Complete plan, sections, sign schedule, quantities and setting-out for both roads.

Standard values used (Oman HDS 2017, Section 19)

Sign plate sizes by speed (mm) and warning distance

Speed	Triangle	Circle	STOP	GIVE WAY	Warning
50	600	600	750	750	50 m
60	750	750	900	750	75 m
80	900	750	1200	900	150 m
100	1200	900	–	–	200 m
120	1500	1200	–	–	300 m

Delineator post spacing on curves (S)

Radius (m)	500	300	200	100	75	50	25
Spacing S (m)	50	30	20	15	12	10	7

Approach tangents 2S, 3S, 6S (max 100 m). Posts 0.30 m beyond the paved shoulder. Sign lower edge 1.5 m above ground (2.1 m over a sidewalk). Warning triangles yellow; chevrons yellow / black permanent, black / yellow temporary; cones red with a white band.

Road markings

Line	Use	Pattern
L1	Lane / centre line	2/6 - 3/9 - 4/12 m
L2	Warning (bends)	6/2 - 9/3 - 12/4 m
L2A	No overtaking	Continuous
L4	Edge (yellow)	Solid, 120 mm
L5	Beside aux lane	1 m / 1 m
L5/3	Across side road	600 / 300 mm
L7 / L8	Stop / give way	Line + STOP / triangle

Please note Values follow the Oman Highway Design Standards 2017, Section 19 (GCC manual based). For projects in Qatar, check against the Qatar Traffic Control Manual (QTCM) and your project specification. Every value can be changed in the dialogs.

Troubleshooting and quick fixes

Problem	Fix
Unknown command RDP	Load RoadPro.lsp with APPLOAD. Full AutoCAD only, not AutoCAD LT.
Road is 1000 times too small or too big	Set Drawing units to Millimeters for mm drawings (Meters for m).
"No RoadPro data on this object"	Select the red centre line on layer ROAD-CL of a road built by RDP or RDJ.
Warning: edge treatment tighter than the radius	Increase the radius, or reduce the width, shoulder or unit width (slide 12).
Black outline not visible	Dark screen background: outlines show white but plot black.
Excel file could not be saved	Close the old CSV in Excel, save the DWG, then run RDR again.
Junction not formed	Start the branch exactly ON the main CL (snap Nearest), away from widened or tapered edges.
Auxiliary lane skipped	With its tapers it would run past the road end: move the junction or shorten the tapers.
Widening runs outside the road	Start – taper in and End + taper out must stay inside the road chainages.
Chainage not accepted	Type 100 or 0+100 as a real road chainage (includes the start chainage).
RDL placed nothing	From / To outside the line length, or spacing larger than the line. Set To = 0 for the line end.
Settings look wrong	Type ROADPRORESET to restore all defaults (width 7.30 m, radius 10 m).
Walkways show herringbone, not the I-shape paver	RP_PAVER.pat could not be written / found: copy it to a Support folder (it is in the package).
RDER refuses a curve	Curves touching another curve, or at the line end, need a straight each side - edit by hand + RDB.
Civil 3D alignment not converted	Use RDLX / LandXML in Civil 3D (Insert > LandXML) and RDLXI, or explode a copy to a polyline.

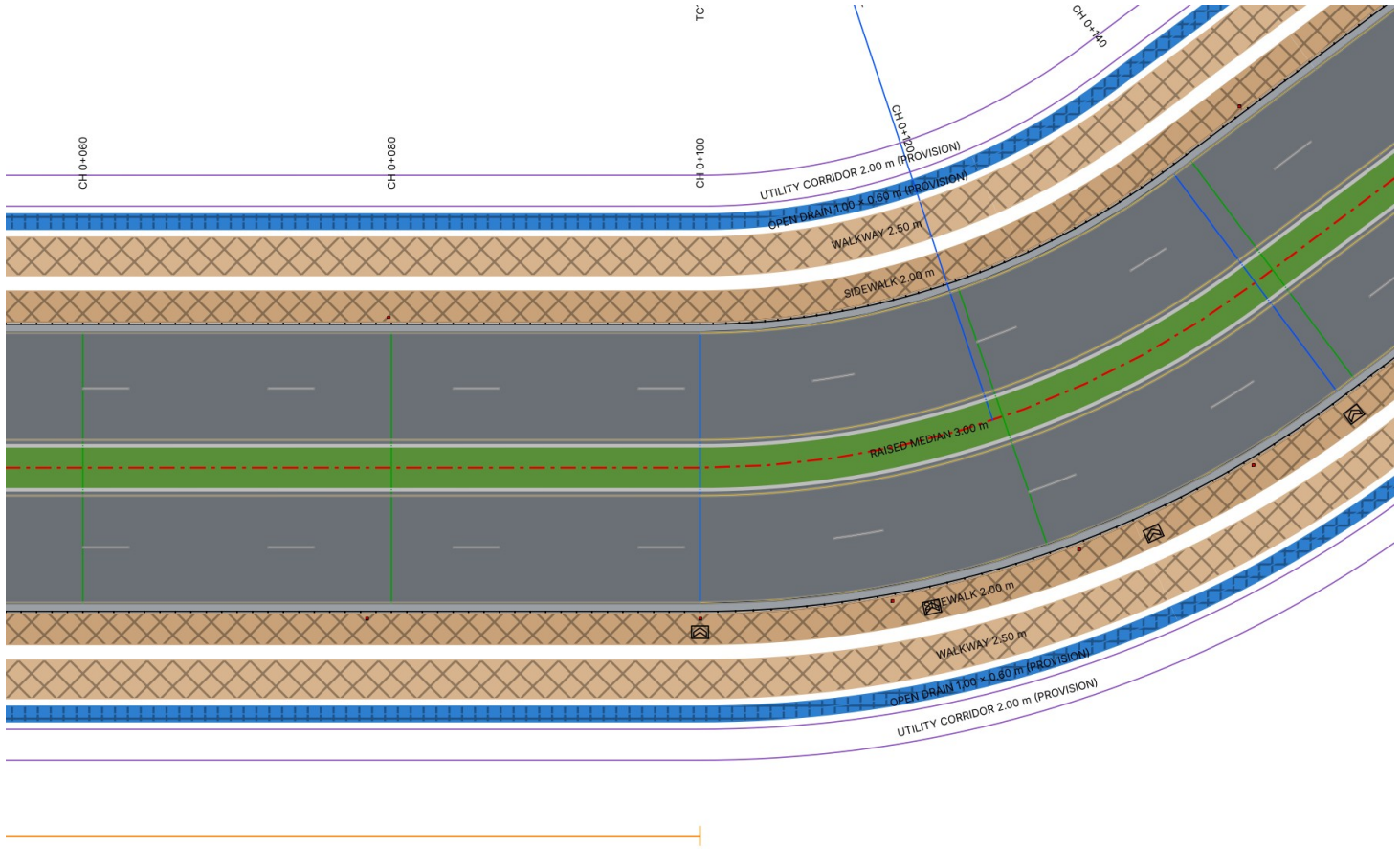
Quick reference: shortcuts and default values

RDP	Build road	RDR	Report + Excel	RDRAD	Radius labels	RDSEC	Straight sections	RDX	Cross sections
RDW	Lane widening	RDT	Road taper	RDJ	Junction / tie-in	RDM	Road markings	RDL	Single line
RDC	Corridor (new)	RDU	Update corridor	RDER	Edit radius (new)	RDB	Rebuild road	RDLX	LandXML out / in

RDP default	Value	RDW / RDT / RDX	Value
Road width	7.30 m	Extra lane width	3.65 m
Manual radius	10.0 m	Widening tapers	30 / 30 m
Unit / cone spacing	2.00 m	Taper edge shift	+1.00 m
Unit width	0.50 m	Fillet radius	30 m
Paved shoulder	0.50 m	Crossfall	2.5 %
Speed	30 km/h	Barricade height	1.00 m
Chainage interval	20 m	Sign clearance	1.50 m

RDJ / RDL default	Value
Branch width	7.30 m
Corner radius	10.0 m
Corner treatment	Same as main
Aux lane / tapers	3.65 / 60 / 30 m
Island	1.5 x 15 m
RDL offset / unit	0.50 / 2.00 m
RDL To distance	0 = line end
Lane width / median	3.65 / 3.00 m
Sidewalk / walkway	2.00 / 2.50 m

The full corridor in one build



Real RoadPro output (simulator render): dual carriageway 2 + 2 lanes, raised median 3 m, sidewalks and walkways with interlocking pavers, open drains, utility corridors.

Lanes
From road width, manual (left / right can differ) or automatic

Carriageway
Single or dual, with lane lines and per-lane arrows

Median
Flush (painted, hatched) or raised (kerbed), any width

Sidewalks
Left, right, both or none - own widths

Walkways
Separate pedestrian walkway beyond a verge

Drainage
Open channel, covered U-channel or pipe (provision)

Utilities
Utility corridor strip each side (provision)

Paver hatch
Interlocking I-shape paver, follows the road

Section + report
Every item in RDX sections and in RDR / Excel

Carriageway and lanes: manual or automatic

Setting	Options (default)	What it does
Carriageway	Single / Dual	Dual = two carriageways separated by a median or a centre barrier.
Number of lanes	From road width / Manual / Automatic	From road width = v3.3 behaviour (no lane lines).
Lanes, left side	1 (manual)	Opposite direction. Traffic keeps right.
Lanes, right side	1 (manual)	Direction of increasing chainage. May differ from the left.
Lane width (m)	3.65	Error below 2.50 / above 5.00; warning outside 3.00 - 3.75.
Design volume per direction	0 = unknown (automatic)	veh/h in the busiest direction.
Lane capacity assumed	1800 veh/h/lane (automatic)	Your project value - it is shown in the report.

Automatic mode

- With volume: lanes = V / C rounded up
- Without: as many lanes as fit in the RDP road width
- Minimum 1 per direction, 2 on a dual carriageway
- Road width = lanes x lane width + median

Example: 2 + 2 lanes x 3.65 m + 2.00 m median = 16.60 m road width. Manual mode fixes the width (the RDP width box is greyed out).



Note Automatic lane counts are indicative only - not an engineering approval. Confirm with a traffic study and your design standard.

Median, unequal lanes and the markings drawn

Flush median (painted)

- Diagonal hatch on ROAD-MEDIAN, white boundary lines
- Width 0.30 m minimum
- Yellow edge line on each side

Raised median (kerbed)

- Kerbs 150 mm on both sides, green fill
- Warning under 1.20 m
- In sections: kerbs with 150 mm upstand

Centre barrier instead

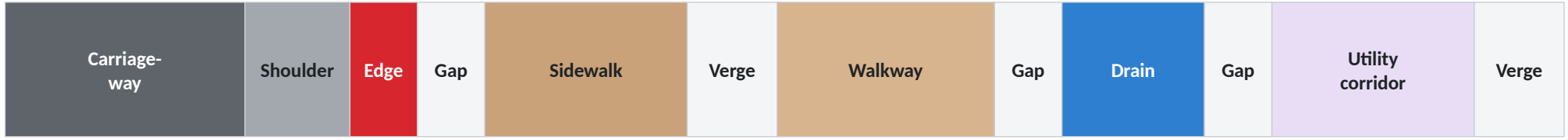
- RDP Centre = barricades or cones
- Allowed as separator of a dual carriageway
- Placed on the median line

Unequal lanes (e.g. 1 left + 2 right)

- The centre line (CL) stays in the middle of the paved road.
- The median / direction line moves by (right - left) x lane width / 2 towards the side with fewer lanes: 1 + 2 lanes x 3.65 -> 1.825 m left.
- Edges, barricades and chainage follow the CL as before.

Marking	Drawn by RDP / RDM
Lane lines	Broken L1 by speed, between lanes
Direction line	L1 / L2 / L2A or double, at the median line
Median edges	Yellow continuous, 0.30 m from the median
Lane arrows	One per lane, in its direction
Widening (RDW)	Edge line follows; L5 line at the old edge

Everything beyond the edge, in a fixed order



Seen from the road edge outwards, each side on its own. Items that are switched off simply close up.

Setting	Default	Notes
Sidewalk left / right + width	Off, 2.00 m	Each side on its own: both, one or none. Warning under 1.50 m.
Gap behind the edge treatment	0.00 m	Space between kerb / barricade and the sidewalk.
Walkway left / right + width	Off, 2.50 m	Side-by-side pedestrian walkway beyond a verge.
Verge before the walkway	1.00 m	Planting / service strip between sidewalk and walkway.
Drain left / right + type	Off, open channel	Open channel, covered U-channel or stormwater pipe.
Width (pipe diameter) / depth / gap	1.00 / 0.60 / 0.50 m	Geometric provision only - no hydraulic design.
Utility corridor left / right	Off, 2.00 m, gap 0.50	Reserved strip for services (provision only).

Follows the road

- Curves, widenings (RDW) and tapers (RDT)
- Gaps at junction mouths (sidewalks, walkways, open drains)
- Pipes and utility corridors run on
- Warning when an item is wider than an inside radius

HOW TO

Build a dual carriageway with sidewalks and walkways

1. **Type RDP**
   Set road name, speed, edge = Kerb stone (width 0.15), Centre line input.

2. **Press Lanes, median, sidewalks, drains, utilities...**
   The corridor dialog opens (bottom of the RDP dialog).

3. **Carriageway = Dual, Number of lanes = Manual**
   Lanes left 2, right 2, lane width 3.65.

4. **Median = Raised (kerbed), width 3.00**
   Or Flush for a painted median.

5. **Tick Sidewalk left / right and Walkway left / right**
   Widths 2.00 / 2.50, verge 1.00. Add drains / utilities if needed.

6. **Check layout, then OK**
   The check line shows the road width (17.60 m) and any warnings.

7. **OK in the RDP dialog, draw or select the CL**
   Road, median, corridor, markings, signs, report and section are drawn.

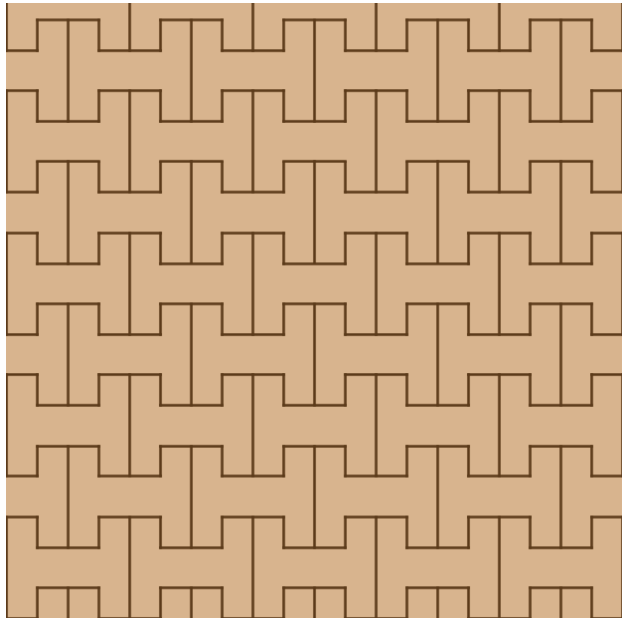
8. **Later changes: RDC on the road**
   Sidewalk / drain changes update only the corridor; lane / median changes rebuild the road.

```
Command: RDP
  Manual lanes: 2 left + 2 right x 3.65 m
  + median 3.00 m = road width 17.60 m
  ...
  [CROSS SECTION / CORRIDOR]
  Median - raised (kerbed) : 3.00 m wide ...
  Sidewalk - left  : 2.00 m wide, area ...
  Walkway - right  : 2.50 m wide, area ...
```

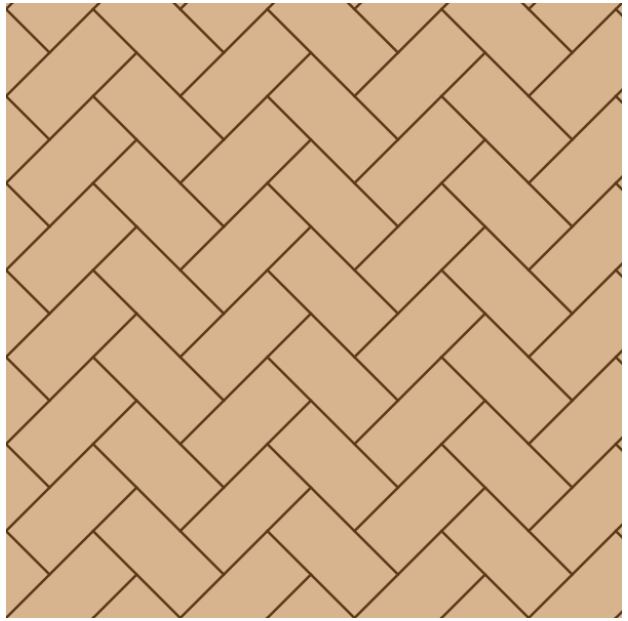
Other commands

RDC change the corridor of an existing road
RDU regenerate corridor + markings (after RDW / RDT / edits)
RDX cross sections now show every item with dimensions
RDR report and Excel list lengths and areas

Interlocking paver hatch for sidewalks and walkways



RP_PAVER - I-shape interlocking paver 200 x 165 mm (default)



AR-HBONE - herringbone, AutoCAD standard (option)

Checked The RP_PAVER definition was drawn line by line from the .pat file and compared with the true I-shape tiling - they match exactly (left image).

Setting	Options (default)
Paver hatch	On (off = solid fill)
Pattern	Interlocking I-shape / Herringbone / Other
Other pattern name	Any AutoCAD pattern, e.g. AR-B816
Scale	1 = real paver size
Extra angle (deg)	0 = along the road

How it works

- RP_PAVER.pat is written once to your roaming Support folder (it is found automatically).
- The hatch is clipped to each sidewalk / walkway piece and turns with the road (25 m pieces on curves).
- If the pattern cannot be loaded, RoadPro uses AR-HBONE and says so on the command line.
- mm drawings: scale x 1000 applied automatically.

Change a curve radius after the road is built

1. **Type RDER**
   Pick the road centre line ON the curve you want to change.

2. **Read the current curve**
   Number, radius (CL), deflection and TC / CT chainage are shown.

3. **Type the new radius at the centre line**
   Or I = radius at the inner edge (kerb line), A = all curves.

4. **RoadPro re-fits the curve**
   Between the same two straights, always tangent to both.

5. **The road is rebuilt**
   Edges, surface, barricades / cones / kerb, markings, signs, labels and corridor - with the road's own settings.

Checks before anything changes

- Radius larger than half the road width
- Fits between the points either side - else the largest possible radius is shown
- Curves that touch another curve (compound / reverse) are not edited
- One U undoes the whole edit

Edited the CL by hand? RDB

Grips, Properties or PEDIT on the red CL, then RDB (ROADREBUILD) draws the road again from it.

Please note Widenings, tapers and junctions are not carried through a rebuild (their chainages move). RoadPro lists them, asks first (Enter = No) and you apply them again with RDW / RDT / RDJ. Report tables and sections drawn before are kept - run RDR / RDX again.

Existing polyline, manual alignment, Civil 3D and LandXML

A Follow an existing line

- RDP, Centre line input = Select existing
- Line, arc, 2D polyline or Civil 3D alignment
- Keep the original line: the road is built on a copy (default on)
- Checks: short segments, sharp bends without a curve, radius under half width, self-intersection (asks before building)

B Manual alignment

- Centre line input = Pick points: PLINE with Arc option
- Manual radius at every bend, at the CL or inner edge
- Change any curve later with RDER, or edit by hand and RDB
- Circular curves; transitions in width with RDW / RDT (no spirals)

Civil 3D and LandXML

RDLX centre line -> LandXML 1.2 alignment (lines, arcs, start station). Civil 3D: Insert > LandXML.

RDLXI LandXML -> polylines (arcs exact, spirals as chords), then RDP.

Select a Civil 3D alignment directly in RDP (copy exploded and joined; start station read).

Limits RoadPro is an AutoLISP plugin, not a Civil 3D corridor. Direct alignment selection depends on Civil 3D exploding a copy of the alignment into lines and arcs - it was not tested in Civil 3D. The LandXML route (RDLX / RDLXI) was tested end to end and is the recommended exchange. Stations, curves and coordinates are kept; profiles and superelevation are not exchanged.

How version 3.4 was tested - and what to check on your PC

105 automated checks passed

Every new feature, integration and end-to-end case

0 differences to 3.3

Plain RDP roads identical; only intended fixes elsewhere

293 functions statically checked

No undefined calls, no argument-count errors

How

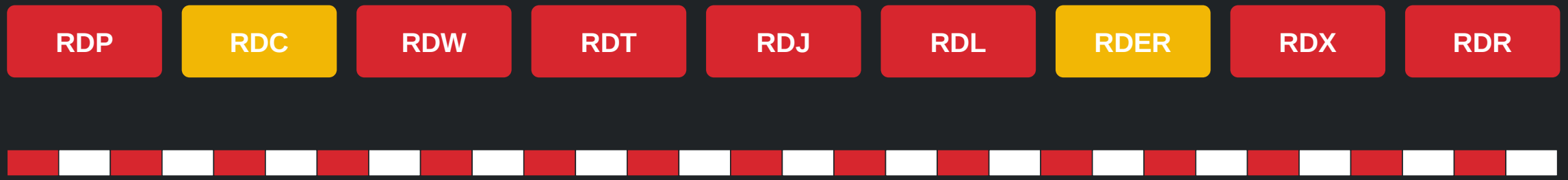
- The real RoadPro.lsp was run in an AutoLISP interpreter with an emulated AutoCAD drawing (geometry, offsets, hatches, xdata, dialogs).
- Results were measured (offsets, radii, tangency, areas, counts) and rendered to images.
- Not AutoCAD itself: display, plotting and Civil 3D behaviour still need a check on your PC.

Please check in AutoCAD / review

- Load, RDP with the Lanes... dialog, RDER, RDLX import in Civil 3D
- RP_PAVER hatch display and plotting
- Direct Civil 3D alignment selection
- Engineering review: lane widths and counts, median and sidewalk widths, drainage and utility sizes (provisions only) against QTCM / QHDM and the project specification

Build your next road with RoadPro

Version 3.4 · 23 October 2026 · RoadPro.lsp for AutoCAD



Developed by https://www.exceltokml.com/

Standards reference: Oman HDS 2017 Section 19. Check QTCM / QHDM and your project specification for Qatar.